JONATHAN JONGBIN KIM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Republic of Korea
B.S. in Electrical Engineering *Mar. 2025 – Present*

GPA: 3.65 / 4.3 | **Double Major:** Computer Science (in progress)
Relevant Coursework (completed and in progress): Introduction to Reinforcement Learning, Introduction to Generative Models, Machine Learning Basics and Practices, Probability and Introductory Random Processes, Programming Languages, Analysis I, Linear Algebra, Data Structures and Algorithms, Signals and Systems

RESEARCH INTERESTS

Foundations of machine learning, learning and reasoning, world models, and reinforcement learning.

MAJOR PROJECTS

dongne-rec — Hyperlocal Sequential Recommendation [GitHub] [Demo] 2026

- Built SASRec and four offline ranking protocols on 618K Foursquare check-ins from New York and Tokyo. In per-user leave-one-out, SASRec led with sampled candidates, while a visit-frequency baseline led on all candidates within 3 km.
- Built a FastAPI inference prototype. ONNX INT8 reduced the encoder from 13.4 to 3.5 MB. Local HTTP benchmarking measured p99 of 8.51 ms at concurrency 4 on an Apple M4 Pro.

DiscoSpatial — Meta-Learned Updates for Spatial Memory [GitHub] 2026

- Built a recurrent RGB Memory Maze benchmark comparing frozen Disco103 with V-trace actor-critic. Used 57 GPU pilot runs, synthetic positive controls, and reward-credit diagnostics to investigate near-uniform policies.

FastGen — Few-Step Image Generation (Team Project) 2026

- Built reproducible teacher–student distillation experiments for one and four model evaluations per image, with fixed data splits and Clean-FID evaluation.
- Compared Euler and DPM++ 2M in a controlled pilot. Trajectory loss fell while training-set FID rose.

3D Gaussian Splatting — Scene Reconstruction [GitHub] 2026

- Implemented a differentiable PyTorch renderer and a COLMAP-to-gsplat training pipeline with densification controls and TensorBoard logging.
- Trained the gsplat pipeline on 135 registered views for 30,000 steps on a Tesla T4, reaching 27.354 dB validation PSNR and 0.03535 L1 error.

OTHER PROJECTS

DiT-VITON — Virtual Try-On [GitHub] 2026
Made DiT latent resolution configurable. Exposed image size across preprocessing, training, and inference, plus dataset and batch size options for training.

StarGAN v2 — Multi-Domain Image Translation [GitHub] 2025
Trained a PyTorch implementation on AFHQ. FID 44.11 and LPIPS diversity 0.4270 on 3,000 samples.

PUBLICATIONS & AWARDS

[1] **J. Kim**, et al., "*Analyzing The Impact and Evaluation of Flextime on Subway Congestion Using SARIMA Model*", *Proceedings of the Korea Computer Congress (KCC)*, 2024.

[2] **J. Kim**, et al., "*Comparison of Deep Learning Architecture Performance and Suggestion of Baseline Model for Diagnosing Cassava Leaf Disease*", *Proceedings of the Korea Software Congress (KSC)*, 2024.

Award Excellence Award, Junior Division, KSC Paper Competition (2024).

SKILLS

Programming: Python, C++, Bash | **ML:** PyTorch, NumPy, ONNX Runtime
Tools: Git, Linux, CUDA, LaTeX